Mechanics of Human Movement

Course Progress 90.2% (11 of 13)

- Week 4
- Week 5
  - Static Analysis of Hip- Part I
  - Static Analysis of Hip-Part II
  - Static Analysis of the Knee
  - Static Analysis of the Knee and Ankle
  - Kinetics: Linear Motion
  - Week 5 - Lecture Notes
  - Mechanics of Human Movement : Week 5 Feedback Form
  - Quiz: Week 5: Assignment 5
- Week 6

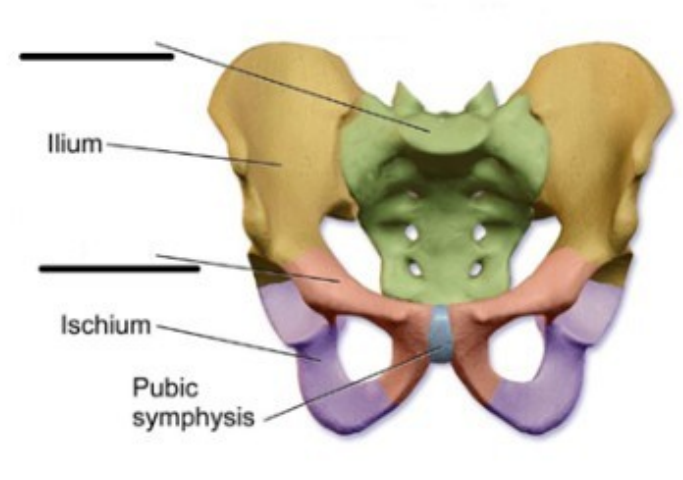
Assessment

Week 5: Assignment 5

Due date on 2026-08-26, 23:59 IST

The due date for submitting this assignment has passed. As per our records you have not submitted this assignment.

1. 1 Point



Mark the missing names of the anatomical parts (posterior to anterior) of the above figure from the list below

○ Sacrum and Pubis

○ Sacrum and Anterior Iliac Spine

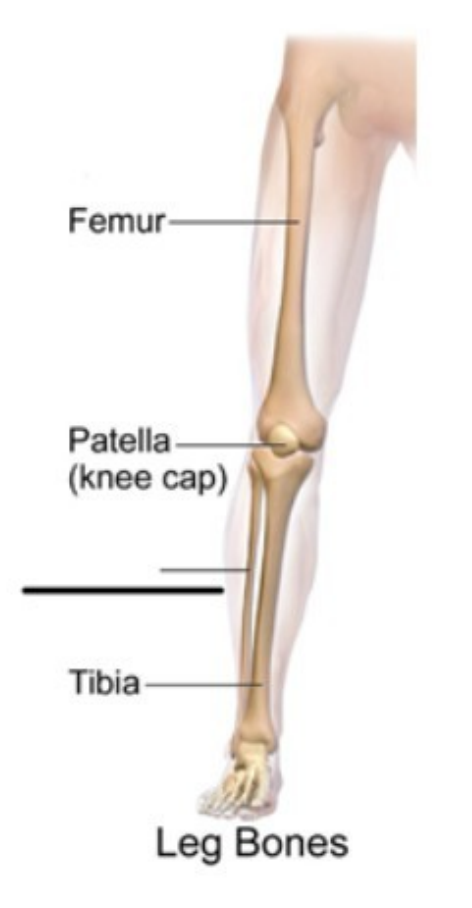
○ Posterior Iliac Spine and Sacrum

○ Pubis and Sacrum

You didn't attempt this question well in time. Score: 0

Accepted answer is Sacrum and Pubis

2. 1 Point



Mark the missing names of the anatomical parts of the above figure from the list below

○ Fibula

○ Humerus

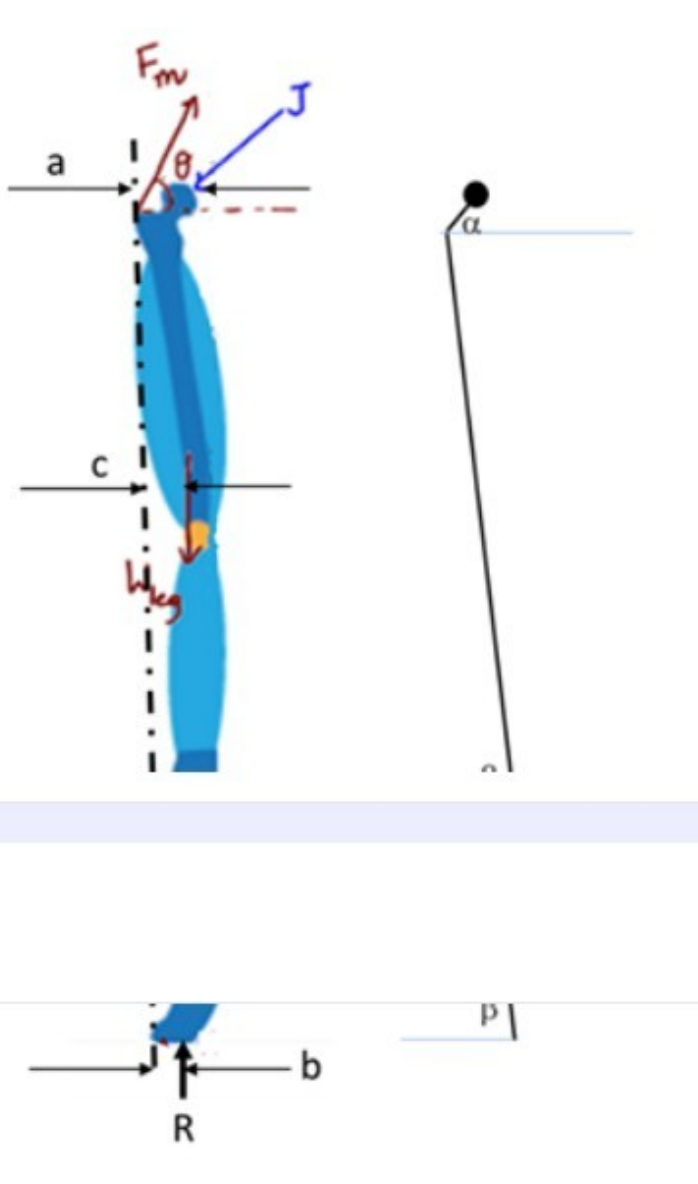
○ Sternum

○ Clavicle

You didn't attempt this question well in time. Score: 0

Accepted answer is Fibula

3. 1 Point



Using the free body diagram of single leg stance shown above to answer questions 3-5. Given: W_{leg} = 0.2W, a=7cm, b=17.8cm, c=10.2cm, β = 80 °, θ=70 °

Calculate F_m, assuming that the angle α made by the femoral neck with the horizontal is 45 °

○ 4.6W

○ 3.2W

○ 1.1W

○ 2.4W

You didn't attempt this question well in time. Score: 0

Accepted answer is 2.4W

4. Calculate the net joint reaction force 1 Point

○ 4.6W

○ 3.2W

○ 1.1W

○ 2.4W

You didn't attempt this question well in time. Score: 0

Accepted answer is 3.2W

5. Calculate the angle made by the net joint reaction force with the horizontal 1 Point

○ 45^0

○ 60^0

○ 75^0

○ 90^0

You didn't attempt this question well in time. Score: 0

Accepted answer is 75^0

6. Carrying a load on one side increases the loads on the hip on the other supporting side in single stance 1 Point

○ True, the joint has to produce the necessary force for static equilibrium

○ False, the supporting side need not balance the external force

○ True, the muscle has to produce the necessary counter moment for static equilibrium, which increases joint force

○ False, the hip does not bear the load but the thigh does

You didn't attempt this question well in time. Score: 0

Accepted answer is True, the muscle has to produce the necessary counter moment for static equilibrium, which increases joint force

7. People with weak hip abductor muscles and/or painful hip joints lean towards the loading side in single stance. 1 Point

○ False, that will increase joint reaction force in their already painful hip joint

○ True, the moment that needs to be balanced decreases, thereby reducing the muscle force required

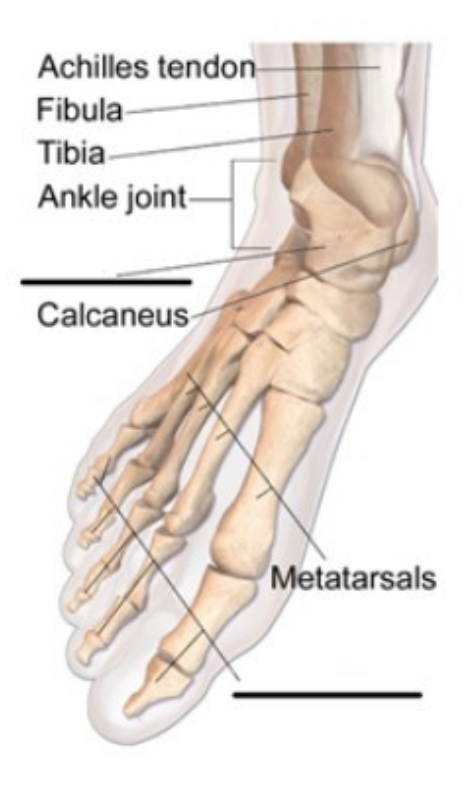
○ False, the iliopsoas would balance for the weak hip abductors and hence they do not have to lean

○ True, such a compensatory gait would help reduce muscle force since it decreases external force in vertical direction

You didn't attempt this question well in time. Score: 0

Accepted answer is True, the moment that needs to be balanced decreases, thereby reducing the muscle force required

8. 1 Point



Match the missing anatomical terms in the above figure from the list below.

☐ Talus

☐ Carpal

☐ Phalange

☐ Pisiform

You didn't attempt this question well in time. Score: 0

Accepted answer is Talus, Phalange

9. Which of the following best describe(s) the function(s) of patella 1 Point

☐ Enhances the leverage of the quadriceps tendon.

☐ Increases the moment generated by quadriceps about the knee joint.

☐ Protects the anterior aspect of the knee joint from physical trauma.

☐ Increases the range of motion of the knee joint.

You didn't attempt this question well in time. Score: 0

Accepted answer is Enhances the leverage of the quadriceps tendon., Increases the moment generated by quadriceps about the knee joint., Protects the anterior aspect of the knee joint from physical trauma.

10. The motion of the upper surfaces of the toes moving towards the shin bone is known as __________. 1 Point

○ Dorsiflexion

○ Plantarflexion

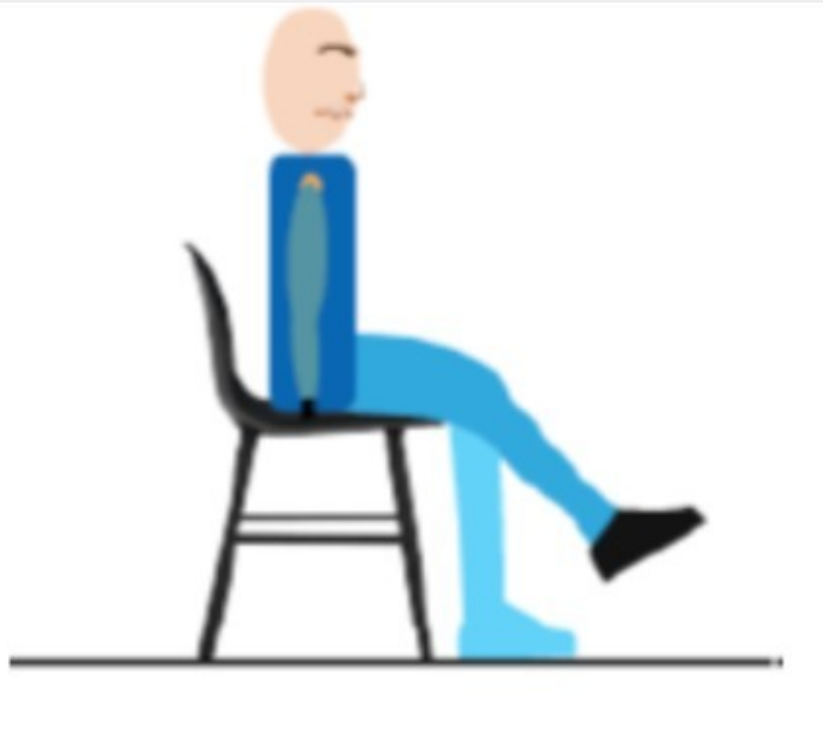
○ Extension

○ Upward rotation

You didn't attempt this question well in time. Score: 0

Accepted answer is Dorsiflexion

11. A person is performing knee strengthening exercises with ankle weights in a quasistatic manner as shown above. Find the force on the patellofemoral joint when the shank is inclined at 30° to the vertical given the following data: the weight of the leg is 150N and an ankle weight of 50N is being used, the patellar tendon inserts on the shank at a distance of 12 cm from the proximal end at an angle of 15° to the axis of the shank, and the quadriceps tendon acts parallel to the thigh. Assume the length of the leg is 50 cm and its weight acts at its mid-pt. 1 Point



○ 525N acting at an angle of 52.5 ° to the horizontal

○ 1006N acting at an angle of 75 ° to the horizontal

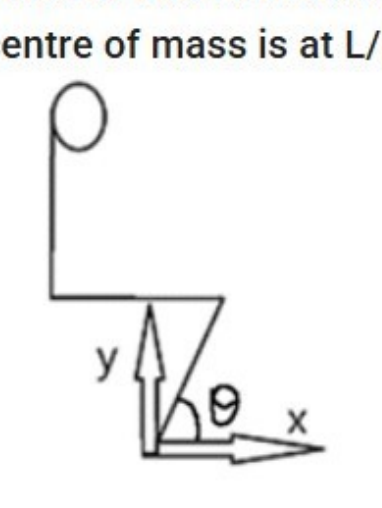
○ 1225N acting at an angle of 52.5 ° to the horizontal

○ 1725N acting at an angle of 60 ° to the horizontal

You didn't attempt this question well in time. Score: 0

Accepted answer is 1225N acting at an angle of 52.5 ° to the horizontal

12. The human body is represented as 3 rods, each of mass m and length L, calculate the x and y coordinates of the centre of mass of the system shown in the figure below. Let the angle made by the shank segment with the x axis be θ. Assume the mass of each link to be uniformly distributed except for the vertical link (equivalent of the head, arms and torso) whose centre of mass is at L/3 from its distal end. 1 Point



○ $\frac{L(5\cos\theta-3)}{6}, \frac{L(2+15\sin\theta)}{18}$

○ $\frac{L(5\cos\theta-3)}{2}, \frac{L(8+15\sin\theta)}{6}$

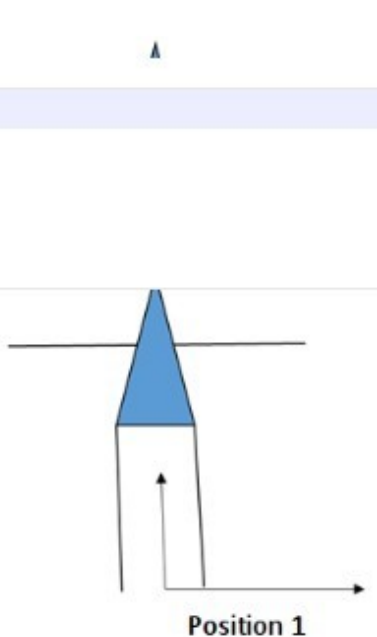
○ $\frac{L(7\cos\theta-4)}{3}, \frac{L(8+9\sin\theta)}{3}$

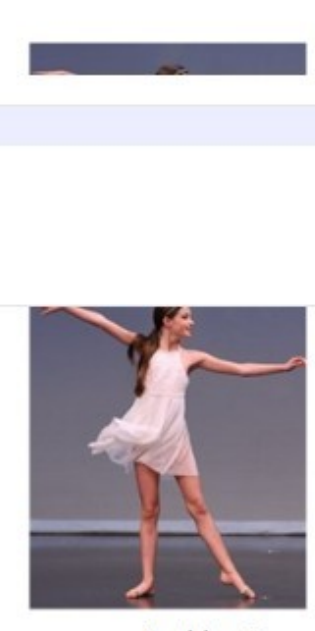
○ $\frac{L(\cos\theta-2)}{3}, \frac{L(7+15\sin\theta)}{3}$

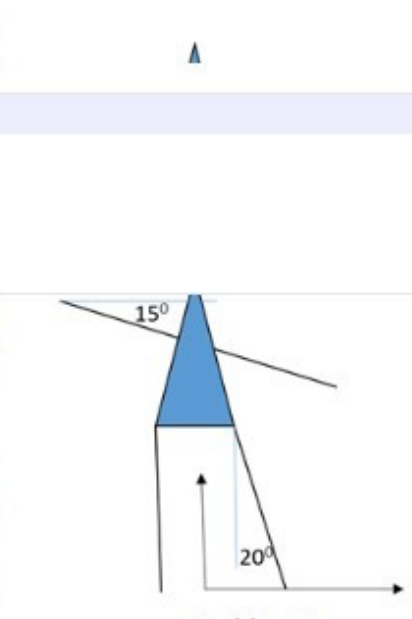
You didn't attempt this question well in time. Score: 0

Accepted answer is $\frac{L(5\cos\theta-3)}{6}, \frac{L(2+15\sin\theta)}{18}$

13. 1 Point


Position 1


Position 2



Position 2

The new position of the COM of her arms alone will be

☐ Higher than and towards the right of her body as compared to the first position

☐ Lower than and towards the left of her body as compared to the first position

☐ Higher than and on the midline of her body as compared to her anatomical position

☐ Unchanged with respect to the COM of her arms in the first position

You didn't attempt this question well in time. Score: 0

Accepted answer is Higher than and on the midline of her body as compared to her anatomical position, Unchanged with respect to the COM of her arms in the first position